

COTTBUS, Germany

WMO: 104960

Lat: 51.7761N

Lon: 14.3169E

Elev: 228

StdP: 14.58

Time Zone: 1.00 (EUC)

Period: 90-03

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	10.7	15.9	6.0	7.5	11.2	11.5	10.0	16.7	24.1	46.5	21.8	45.6	5.4	70	0.511

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
8	20.9	88.1	66.9	84.6	65.3	81.6	64.2	69.2	84.6	67.4	80.3	65.7	77.6	6.7	250

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
64.4	91.2	72.8	62.7	86.0	71.7	61.1	81.1	70.9	33.3	85.1	31.9	80.3	30.6	77.7	74.1

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 19.9	16.8	14.5	DB	2.6	96.6	4.9	3.4	-0.9	99.0	-3.7	101.0	-6.5	102.8	-10.0	105.3	(4)
(5)			WB	3.1	72.5	5.2	0.9	-0.6	73.1	-3.6	73.6	-6.5	74.2	-10.3	74.8	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	49.8	33.2	36.8	41.2	48.7	57.5	63.2	66.9	67.7	57.9	49.9	39.9	33.3	(6)	
	DBStd	14.26	8.81	9.34	6.56	7.93	7.18	6.49	5.50	5.80	5.36	7.58	6.64	8.37	(7)	
	HDD50	2237	522	374	282	116	15	0	0	0	3	98	309	518	(8)	
	HDD65	5871	986	789	737	492	252	110	42	39	220	468	754	982	(9)	
	CDD50	2152	1	5	10	78	246	397	525	548	241	96	4	1	(10)	
	CDD65	311	0	0	0	4	18	57	102	122	8	0	0	0	(11)	
	CDH74	3337	0	0	0	61	296	629	1025	1222	101	3	0	0	(12)	
	CDH80	1038	0	0	0	7	69	203	333	414	12	0	0	0	(13)	
(14) Wind	WSAvg	6.2	7.0	7.9	7.4	6.0	5.5	5.6	5.4	5.0	5.6	6.3	6.5	6.4	(14)	
(15) Precipitation	PrecAvg	23.5	1.8	1.5	1.6	1.3	2.2	2.2	3.2	2.5	1.9	1.5	1.9	1.9	(15)	
	PrecMax	35.2	3.0	3.1	3.8	3.5	4.8	5.0	7.6	8.8	4.8	3.5	5.4	3.9	(16)	
	PrecMin	15.5	0.1	0.2	0.6	0.1	0.4	0.4	0.3	0.4	0.4	0.3	0.1	0.6	(17)	
	PrecStd	4.3	0.7	0.7	0.9	0.7	1.2	1.1	1.8	1.7	1.1	0.8	1.1	0.7	(18)	
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	54.5	60.7	66.4	80.9	86.0	91.8	93.9	93.0	81.5	73.8	58.2	54.6	(19)	
		MCWB	48.5	50.7	53.2	60.0	63.3	69.4	70.0	69.7	63.5	62.1	52.1	49.8	(20)	
	2%	DB	50.6	56.3	60.6	73.9	81.6	85.2	87.5	87.7	76.7	69.0	54.8	50.5	(21)	
		MCWB	46.1	48.5	50.3	58.2	62.5	65.5	67.0	67.1	61.4	59.3	49.2	46.7	(22)	
	5%	DB	47.7	52.9	55.5	67.7	76.8	81.1	83.4	84.4	72.8	64.6	52.2	47.6	(23)	
		MCWB	43.5	46.1	48.2	54.8	59.9	64.0	65.2	65.5	61.4	57.0	48.0	44.0	(24)	
	10%	DB	44.7	49.8	51.8	62.7	72.7	76.9	79.6	81.2	68.4	61.1	49.3	44.3	(25)	
		MCWB	41.0	44.4	45.7	51.9	58.2	62.2	63.9	64.8	59.4	54.7	45.9	41.4	(26)	
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	50.0	51.7	54.2	61.4	65.9	71.0	71.5	70.9	66.1	63.0	53.5	49.5	(27)	
		MCDB	53.9	59.3	63.5	77.4	80.3	86.5	89.0	85.7	76.3	71.8	58.6	52.5	(28)	
	2%	WB	46.6	49.5	51.0	58.9	63.7	67.8	68.9	69.2	64.2	60.0	50.0	47.0	(29)	
		MCDB	50.3	55.8	58.5	74.1	78.0	80.9	84.1	84.6	73.3	66.8	53.5	50.4	(30)	
	5%	WB	43.7	46.6	48.8	55.6	61.6	65.5	66.8	67.5	62.2	57.7	48.1	44.3	(31)	
		MCDB	47.3	51.7	54.8	65.8	73.8	77.8	79.1	80.7	69.7	63.2	51.5	47.3	(32)	
	10%	WB	41.2	44.6	46.5	52.9	59.2	63.5	65.2	65.9	60.0	55.6	46.1	41.5	(33)	
		MCDB	44.3	49.3	50.8	61.4	69.8	74.5	76.8	77.5	67.4	60.4	49.3	44.3	(34)	
(35) Mean Daily Temperature Range	5% DB	MDBR	9.7	12.5	14.4	18.7	20.7	19.3	19.4	20.9	17.0	14.8	10.1	8.8	(35)	
		MCDBR	12.4	16.3	21.4	25.9	28.9	27.1	27.3	27.1	24.3	20.1	12.3	11.9	(36)	
	5% WB	MCWBR	9.6	11.0	13.6	13.2	13.9	11.7	11.4	11.2	12.7	12.2	9.0	9.6	(37)	
		MCWBR	12.3	14.1	19.4	23.2	25.2	23.5	23.1	23.3	19.8	17.8	12.0	11.9	(38)	
(40) Clear-Sky Solar Irradiance	taub		0.322	0.347	0.389	0.421	0.420	0.416	0.435	0.429	0.389	0.372	0.347	0.315	(40)	
		taud	2.335	2.314	2.217	2.193	2.243	2.283	2.236	2.272	2.354	2.365	2.370	2.352	(41)	
	Ebn at Noon		214	242	252	258	265	266	258	253	251	229	202	196	(42)	
		Edh at Noon	20	27	36	41	41	40	41	38	31	26	20	17	(43)	
(44) All-Sky Solar Radiation	RadAvg		225	443	772	1280	1560	1657	1571	1372	988	562	260	166	(44)	
	RadStd		32	81	100	176	183	131	195	128	129	74	40	19	(45)	

Historical Trends

		DBAvg	Heating			Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP		HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	
(46) Station Only		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	
(47) Regional (43 neighbors)		+0.72	N/A	N/A	N/A	N/A	+0.68	N/A	N/A	+103	N/A	

Nomenclature: See separate page